Evidently, then, every real number x_0 is an eigenvalue of **X**, and the corresponding eigenvectors are functions (we often call them *eigenfunctions*) that are infinitely concentrated at $x = x_0$. The meaning of this is clear: the wave functions

$$\psi(x) = \delta(x - x_0)$$

represent states in which the particle is located right at the point x_0 on the x axis.

It of course makes a lot of sense that the wave function representing a particle known to be at x_0 is zero everywhere except at x_0. How could it be otherwise? But it is gratifying to see the mathematics confirm this intuition.

Consider the inner product of a state $|\Psi\rangle$ and a position eigenstate $|x_0\rangle$:

$$\langle x_0|\Psi\rangle.$$

Using Eq. 8.2, we get

$$\langle x_0|\Psi\rangle = \int_{-\infty}^{\infty} \delta(x - x_0)\psi(x).$$

By the definition of delta functions given in Eq. 8.4, this integral evaluates to

$$\langle x_0|\Psi\rangle = \psi(x_0). \qquad (8.12)$$

Because this is true for any x_0, we can drop the subscript and write the general equation

$$\langle x|\Psi\rangle = \psi(x). \qquad (8.13)$$